Lab 2 Report

Proportional Control



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1 System Modelling

1.1 Scope and Assumptions

This report identifies the Kitticopter altitude plant and evaluates its three-op-amp proportional controller. The DAQ interface connects the physical circuit to a simulated plant. The required outcomes are tracking accuracy above 92%, overshoot below 30%, and robustness to ±10% aerodynamic and component uncertainty [1].

The laboratory model uses unit mass, constant gravity and drag proportional to vertical velocity. Upward altitude is positive; thrust is linearised around hover. The experimentally unknown quantities are velocity gain A, time constant τ and sensor coefficient k_s.

1.2 Forces and Differential Equation

Figure 1.1 defines the force directions. For upward motion, thrust opposes gravity and drag; drag reverses when velocity reverses.

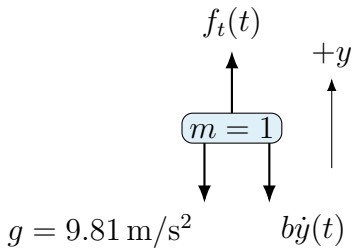


Figure 1.1: Forces on the mass-normalised Kitticopter.

Newton's second law and the local thrust relation give

$$\ddot{y} = f_t - b\dot{y} - g + d_a, \qquad f_t = g + k_T(u - u_h), \tag{1.1}$$

$$\ddot{y} + b\dot{y} = k_T\delta u + d_a, \qquad \delta u = u - u_h, \quad u_h = 2.5\,\mathrm{V}. \tag{1.2}$$

Here y is altitude [m], u is command [V], b is drag [s^{-1}], k_T is thrust slope [$\mathrm{m/(s^2\ V)}$] and d_a is acceleration disturbance [$\mathrm{m/s^2}$]. Gravity cancels at the correctly calibrated hover point.

1.3 Transfer Functions

With zero initial conditions and no disturbance, taking Laplace transforms yields

$$\frac{V(s)}{\Delta F_t(s)} = \frac{1}{s+b}, \qquad G_v(s) = \frac{V(s)}{\Delta U(s)} = \frac{k_T}{s+b} = \frac{A}{\tau s + 1}, \tag{1.3}$$

$$G_y(s) = \frac{Y(s)}{\Delta U(s)} = \frac{A}{s(\tau s + 1)}, \qquad \tau = 1/b, \quad A = k_T/b. \tag{1.4}$$

The position integrator explains why a constant command above hover produces continually increasing altitude rather than a settled position. Differentiating altitude therefore exposes a first-order response suitable for identification.

1.4 Complete Signal Architecture

Figure 1.2 distinguishes the position reference from the voltage command. Both reference and altitude are scaled by k_s [V/m]; the controller acts on their voltage difference.

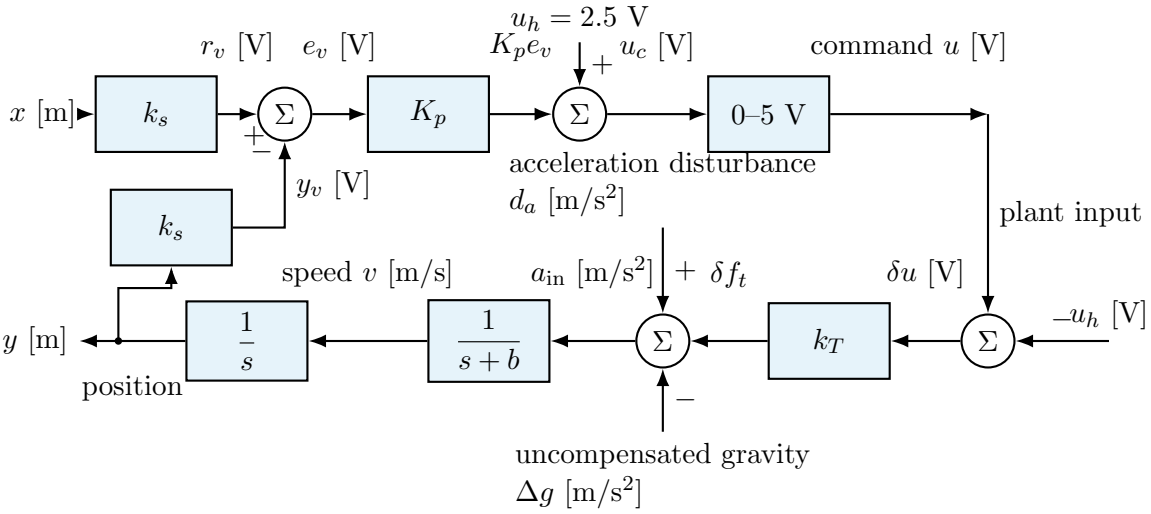


Figure 1.2: Feedback architecture with explicit inputs, speed and position outputs. The lower row flows right to left. Hover cancels nominal gravity; Δg is the uncompensated part.

The analogue law is

$$u = \mathrm{sat}_{[0,5]}\{2.5 + K_p(r_v - y_v)\}, \quad r_v = k_s x, \quad y_v = k_s y. \tag{1.5}$$

The command range is 0–5 V, sensor range 0–10 V and op-amp supply ±15 V. Hover is removed before the deviation plant.

1.5 Closed Loop, Error and Disturbances

In the unsaturated region, define $q = AK_pk_s$ and $D(s) = \tau s^2 + s + q$. Substituting $\Delta U = K_pk_s(X - Y)$ into Equation (1.4) gives

$$L(s) = K_pk_sG_y(s), \qquad \frac{Y(s)}{X(s)} = \frac{L}{1+L} = \frac{q}{D(s)}, \tag{1.6}$$

$$\frac{E_y(s)}{X(s)} = \frac{\tau s^2 + s}{D(s)}, \qquad E_y = X - Y, \tag{1.7}$$

$$\frac{Y(s)}{D_a(s)} = \frac{\tau}{D(s)}, \qquad \frac{Y(s)}{\Delta G(s)} = -\frac{\tau}{D(s)}. \tag{1.8}$$

Uncompensated gravity acts downward. For $K_p > 0$ the loop is stable and Type 1, with zero ideal step error. A constant actuator bias causes a position offset:

$$y_\infty - x_\infty = \frac{d_0}{k_T K_p k_s} = \frac{\delta u_b}{K_p k_s}. \tag{1.9}$$

At 5 m, $K_p = 0.01432$ and an 8% error allowance require $|\delta u_b| < 5.327$ mV: hover calibration and matched scaling are essential.

2 System Identification

2.1 Step-Test Circuit and Recorded Input

Figure 2.1 shows the Session 1 identification circuit and measured input. The 10 kΩ potentiometer wiper feeds the DAQ ADC; the software applies its logged command to the plant. Session 2 instead uses the complete controller circuit.

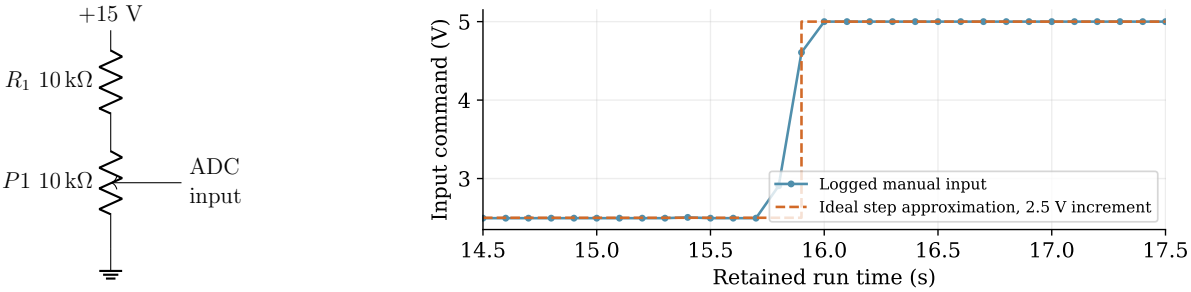


Figure 2.1: Identification circuit (left) and logged input transition (right). Original circuit; Python, MATLAB, data.

The command changes from hover near 2.5 V through 2.908 V at 15.8 s and 4.607 V at 15.9 s to 5.000 V at 16.0 s. The increment is $\Delta u = 2.5$ V. Its short manual transition is retained and checked against an ideal-step approximation, since $\tau \approx 8.1$ s.

2.2 Data Preparation

The unmodified CSV has 1,634 rows. A clock reset separates 31 preliminary rows from the retained 1,603 samples (0.1–160.3 s at 0.1 s spacing). This is one trial; the reattached CSV is byte-identical. The 10 V sensor clips, so velocity uses the unclipped metre channel.

2.3 Parameter Calculations

A backward difference removes the position integrator:

$$v_i = \frac{y_i - y_{i-1}}{t_i - t_{i-1}}. \tag{2.1}$$

For example, $(2.703 - 2.256)/0.1 = 4.47$ m/s at the 16.95 s interval midpoint. The 404 tail differences with endpoint time $t \geq 120$ s average to 35.68748 m/s, giving

$$A = \frac{v_\infty - v_0}{\Delta u} = \frac{35.68748}{5.0 - 2.5} = 14.27499\,\mathrm{m/(s\,V)}. \tag{2.2}$$

Sensor calibration uses 22 nonzero, unsaturated pairs ($0 < y_v < 9.9$ V):

$$k_s = \frac{\sum_i y_i y_{v,i}}{\sum_i y_i^2} = 0.9300026\,\mathrm{V/m}. \tag{2.3}$$

The ratio $2.513/2.703 = 0.92971$ V/m agrees with this regression.

2.4 Time Constant, Sensor and Validation

For a first-order step, $v_{63} = 0.632 v_\infty = 22.55448$ m/s. Interpolating between 22.53 m/s at 24.0 s and 22.69 m/s at 24.1 s gives

$$t_{63} = 24.0 + \frac{22.55448 - 22.53}{22.69 - 22.53}(0.1) = 24.01530\,\text{s}, \tag{2.4}$$

$$\tau_{63} \approx 24.01530 - 15.9 = 8.11530\,\text{s}. \tag{2.5}$$

An exponential fit over 17–60 s gives $\tau = 8.10393$ s and onset $t_0 = 15.91584$ s. Midpoint timing shifts this onset without changing τ.

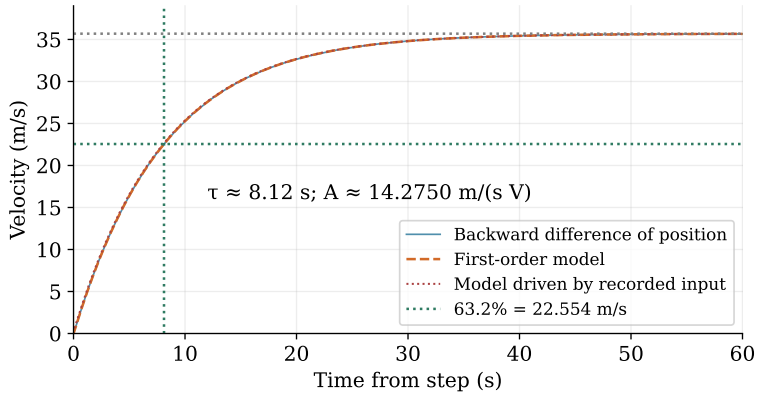


Figure 2.2: Differentiated altitude, adopted first-order model and model driven by the recorded finite input transition. Python, MATLAB, data.

Adopting the submitted values gives

$$G_v(s) = \frac{14.2748}{8.1s + 1}, \quad G_y(s) = \frac{14.2748}{s(8.1s + 1)}, \quad k_s = 0.93. \tag{2.6}$$

Thus $b = 1/8.1 = 0.1234568\ \text{s}^{-1}$ and $k_T = 14.2748/8.1 = 1.762321\ \text{m/(s}^2\ \text{V)}$. Full substitutions and retained samples are in the worked calculation supplement.

2.5 Model Validation and Parameter Check

The rounded model is checked at $t > 60$ s, after fitting 17–60 s. Sensor, residual and integrated-position plots are in the figure folder. The hold-out uses the same trial, so independent validation remains desirable.

Table 2.1: Model validation and parameter-check scores.

Quantity	Result
Velocity RMSE, fit / held-out interval	0.00533 / 0.00436 m/s
Held-out position RMSE	1.686 m
Measured-input velocity RMSE, 17–60 s / > 60 s	0.08114 / 0.00436 m/s
Submitted $A = 14.2748$ / $\tau = 8.1$ / $k_s = 0.93$	100% / 96% / 100%

Integrating $\dot{v} = [A(u(t) - 2.5) - v]/\tau$ from $v(15.7) = 0$ gives transient RMSE 0.23% of final velocity. The laboratory scores in Figure A.1, Appendix A, are not confidence intervals.

3 Controller Design

3.1 Root-Locus Gain Selection

With sensor scaling included, the characteristic equation is

$$1 + K_p k_s G_y(s) = 0 \quad \Rightarrow \quad 8.1s^2 + s + 13.275564K_p = 0. \tag{3.1}$$

The open-loop poles are 0 and $-1/8.1 = -0.1234568$, with no finite zeros. The real-axis locus lies between these poles. Its asymptotes are $\pm 90°$, with centroid and breakaway at $s_b = -1/(2\tau) = -0.0617284$; the breakaway gain is 0.00232489. Matching to $s^2 + 2\zeta\omega_n s + \omega_n^2$ gives

$$\omega_n = \sqrt{AK_p k_s/\tau}, \quad \zeta = \frac{1}{2\sqrt{\tau AK_p k_s}}, \quad \sigma = \frac{1}{2\tau}, \tag{3.2}$$

$$M_p = 100e^{-\pi\zeta/\sqrt{1-\zeta^2}} < 30\% \Rightarrow \zeta > 0.357857, \tag{3.3}$$

$$K_p < \frac{1}{4\tau A k_s \zeta_{\min}^2} = 0.0181544. \tag{3.4}$$

The implemented gain 0.01432 satisfies the nominal bound. Figure 3.1 checks the selected poles; Appendix C gives the magnitude check, and the MATLAB design folder contains the root-locus method.

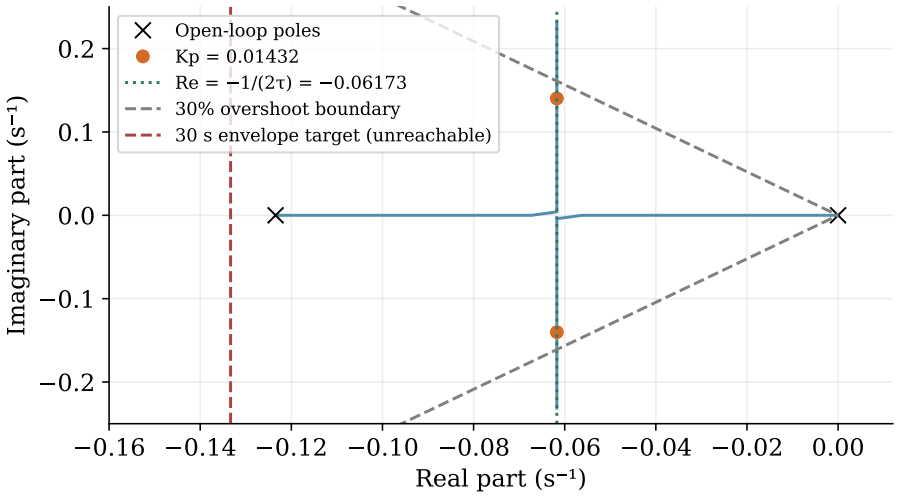


Figure 3.1: Root locus and selected gain. The complex branches have fixed real part, limiting achievable settling speed. Python, MATLAB, data.

Table 3.1: Manual predictions at $K_p = 0.01432$.

Quantity	Value
ω_n / ζ / ω_d	0.153199 / 0.402930 / 0.140212
Closed-loop poles [s^{-1}]	$-0.0617284 \pm j0.140212$
Overshoot / peak altitude for 5 m	25.0804% / 6.25402 m
Peak time π/ω_d / actual 2% settling	22.406 s / 54.89 s

Proportional gain cannot meet the earlier 30 s decay-envelope target because σ is fixed. The $4/\sigma = 64.8$ s approximation is conservative: actual 2% reference settling is 54.89 s.

3.2 Original Simulink Model and Simulation Tests

Figure 3.2 is the original supplied model. Its 0.93 gains scale reference and altitude, 0.01432 is the proportional gain, and the two 2.5 V terms add and remove hover. The velocity lag is followed by the altitude integrator.

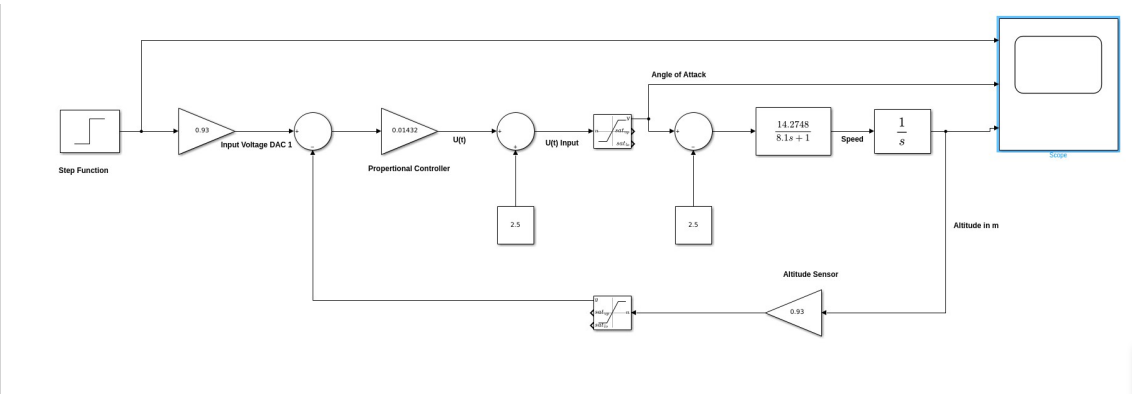


Figure 3.2: Original Simulink model. Native model-building and test scripts.

The supplied scope starts from zero altitude with a 2.5 m reference, changing to 5 m at about 2.5 s. This differs from the isolated 0-to-5 m design test. Figure 3.3 compares the original trace with Python using the same startup sequence. The original scope image is retained in Figure B.2, Appendix B.

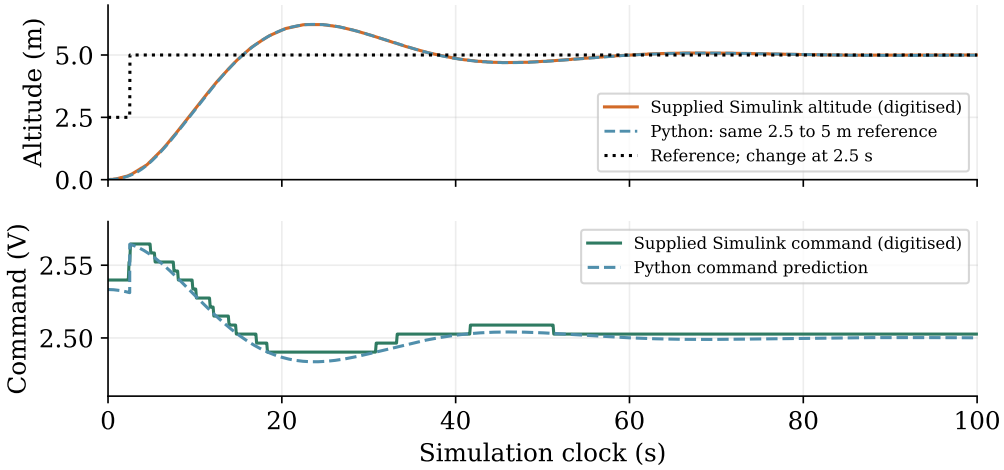


Figure 3.3: Supplied Simulink altitude and command, replotted from the scope and compared with the matching Python prediction. Python, MATLAB, data.

The supplied altitude peaks near 6.226 m (24.51% above 5 m); the matched prediction peaks at 6.231 m. Their altitude RMSE is 0.0115 m, close to the 0.0124 m per-pixel extraction scale. The reconstructed maximum command is 2.56441 V. For the isolated 5 m step, the manual prediction is

$$y = 5\left[1 - e^{-0.0617284t}\{\cos(0.140212t) + 0.44025\sin(0.140212t)\}\right]. \tag{3.5}$$

It predicts 25.08% overshoot and command range 2.48330–2.56659 V. The reconstruction uses 0–5 V command and 0–10 V sensor limits, zero initial states and 100 s duration. Saturation dialogs and the original native model file were not supplied.

3.3 Physical Controller Schematic

Figure 3.4 enlarges the KiCad stages; panels share V1 and V2. The bias/DAQ view is in Appendix C and the board photograph is Figure A.2.

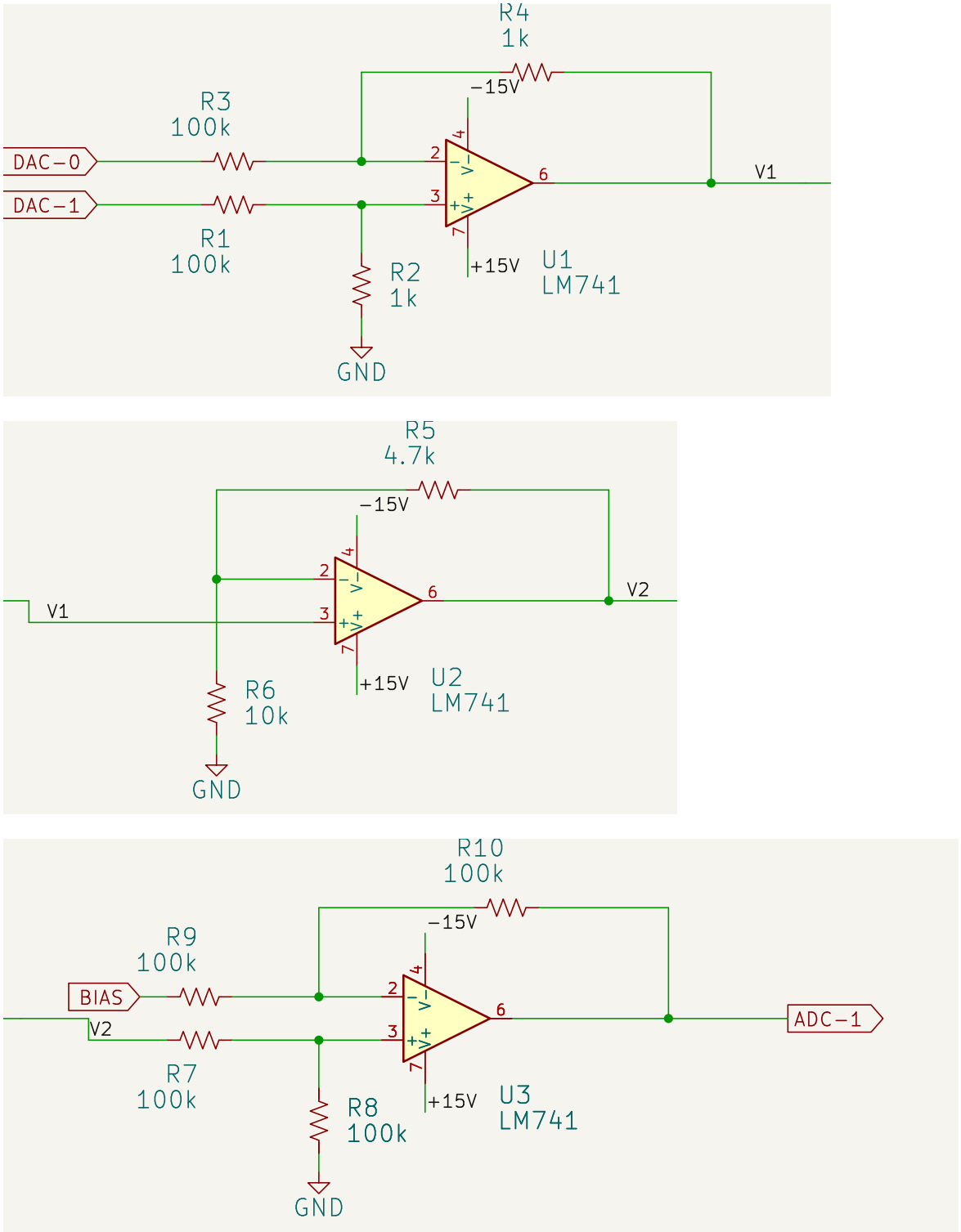


Figure 3.4: Enlarged KiCad stages: U1 attenuates the reference–sensor difference; U2 amplifies it; U3 adds loaded hover bias. $K_p = 0.01(1.47)(0.97428) = 0.0143219$. Editable full schematic and BOM.

4 Controller Tests

4.1 Connected Test and Potentiometer Adjustment

The complete circuit was connected: DAC-1 reference, DAC-0 sensor, ADC-1 command and ADC-2 sensor monitor, with common-ground ±15 V rails. The potentiometer was adjusted until the input was approximately 2.565 V to improve tracking and minimise overshoot; its measurement node and time were not logged. The observation is close to the predicted initial command 2.56659 V; it differs from the 2.500 V zero-error bias and terminal command 2.498 V (Appendix C).

4.2 Experimental Response and Comparison

The required tracking screenshot (Figure B.1, Appendix B) records the demonstration, which received full marks (20/20). At 86.10 s, reference is 5.033 m and height is 4.897 m. Figure 4.1 compares the digitised response with its model.

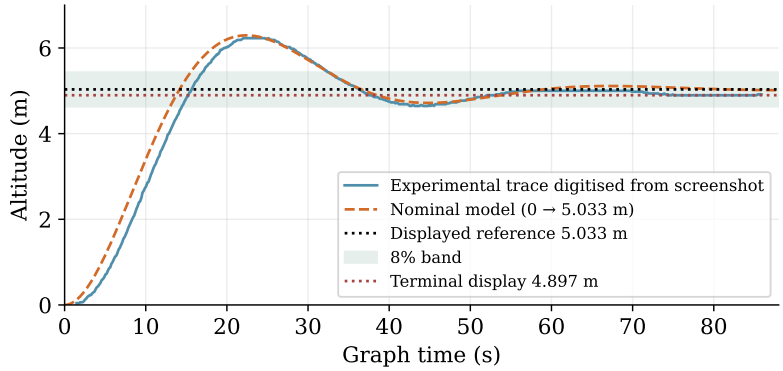


Figure 4.1: Connected controller response replotted with Python and compared with the nominal prediction. Python, MATLAB, data.

The approximate peak is 6.229 m at 22.02 s. The terminal error and reference-based performance are

$$e_f = 5.033 - 4.897 = 0.136\,\text{m}, \tag{4.1}$$

$$\eta = 100(1 - 0.136/5.033) = 97.30\%, \quad M_{p,\text{ref}} = 100(6.229 - 5.033)/5.033 = 23.77\%. \tag{4.2}$$

Table 4.1: Simulation and connected-controller comparison.

Metric	Isolated 5 m model	Original Simulink	Connected circuit
Reference sequence	0 to 5 m	2.5 to 5 m	Near 0 to 5.033 m
Peak altitude	6.254 m	≈ 6.226 m	≈ 6.229 m
Peak above reference	25.08%	≈ 24.51%	≈ 23.77%
Terminal tracking	100% ideal	≈ 99.98%	97.30%

Terminal height passes the 8% band but misses the 2% band, so physical 2% settling is unconfirmed. Overshoot normalised to terminal height is 27.20%, also below 30%.

4.3 Evaluation Against Specifications

Table 4.2 distinguishes observed nominal performance from tolerance predictions. Figure 4.2 varies drag by ±10% while holding the identified thrust slope fixed.

Table 4.2: Specification evaluation and limitations.

Requirement	Evidence	Outcome
Tracking > 92%	97.30% in the retained connected test.	Nominal pass
Overshoot < 30%	Model 25.08%; circuit ≈ 23.77% about reference.	Nominal pass
Drag uncertainty ±10%	Predicted overshoot 21.15–29.45%.	Drag-only pass
Components ±10%	2,048 corners: maximum error 48.79%, overshoot 38.95%.	Not guaranteed

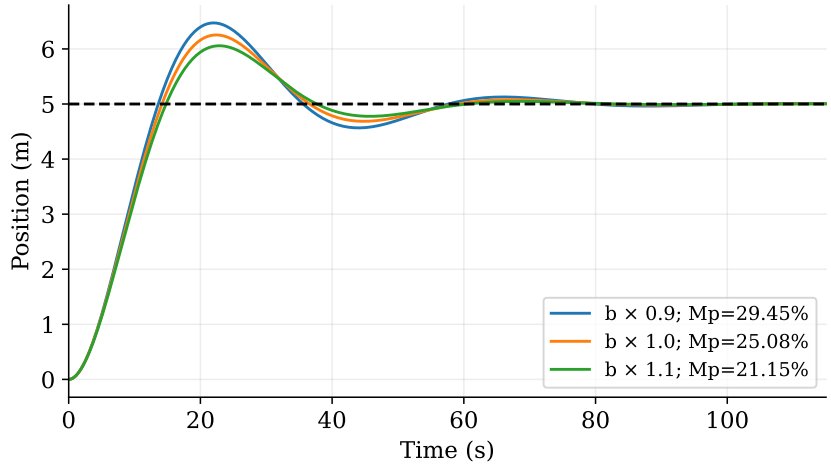


Figure 4.2: Responses for nominal drag and ±10% drag uncertainty, at unchanged thrust slope and gain. Python, MATLAB, data.

For joint scalar drag/gain uncertainty the admissible nominal gain is

$$K_{p,0} < \frac{(0.9b)^2}{4k_T k_s \zeta_{\min}^2 (1.1)} = 0.0133683. \tag{4.3}$$

The implemented 0.01432 exceeds this stricter bound, giving 31.42% worst-case scalar overshoot. Independent resistor mismatch can also change the DC ratio $y_\infty/x_\infty = a_r/a_f$; nominal type number does not remove that error. Appendix C gives the coefficient calculations.

4.4 Conclusion and Recommendations

The controller meets nominal tracking and overshoot requirements, consistent with the full demo mark. Drag-only predictions pass; independent component tolerances and experimental repeatability remain limitations. Use matched reference/feedback ratios, reduce both U1 1 kΩ resistors to matched 680 Ω values for $K_p \approx 0.009739$, and recalibrate the loaded hover output. The corrected scalar worst-case overshoot is 23.44%; the 5 m bias allowance becomes 3.623 mV. These changes require physical retesting. Future tests should retain repeated logs and a defined disturbance record.

References

[1] University of Cape Town, Department of Electrical Engineering, *EEE3094S Lab 2: Proportional Control*, 2026. Supplied laboratory brief and rubrics.

[2] UCT, *EEE3094S Lab 2/3 LaTeX Example*, supplied template. N. Mathebula, *Lab 2 Report*, 2025, supplied structure example.

[3] MathWorks, *stepinfo*. Online documentation.

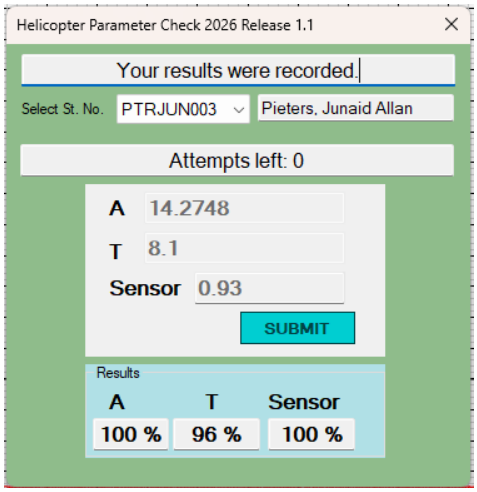
[4] MathWorks, *rlocus*. Online documentation.

[5] MathWorks, *exportgraphics*. Vector PDF and image export documentation.

[6] MathWorks, *sim* and *To Workspace*. Simulation execution; logged output documentation.

Appendix A: Parameter Check and Final Board

Figure A.1 records the submitted parameters and the laboratory check results. The displayed scores support the identification values used in Section 2; they are not uncertainty intervals.



Parameter	Submitted	Score
A	14.2748	100%
τ	8.1 s	96%
k_s	0.93 V/m	100%

The check reports "Your results were recorded" for PTRJUN003. The numerical values agree with the model in the original Simulink diagram. The source image and identification circuit reference are retained in the evidence folder.

Figure A.1: Original Helicopter Parameter Check result for the adopted model.

Figure A.2 documents the final Veroboard controller described in Section 3.3. The three op-amp packages and adjustable bias trimmer are visible. The schematic records the intended connections; a top-side photograph alone does not establish underside continuity.

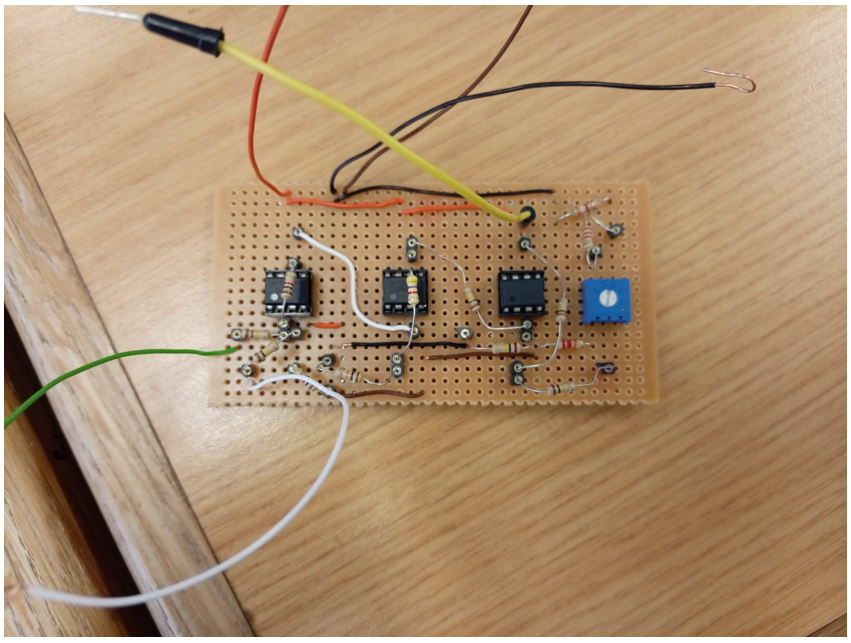


Figure A.2: Final physical controller on Veroboard. The enlarged circuit is Figure 3.4; the editable design is in the KiCad folder.

Appendix B: Demo Record and Original Simulink Scope

The controller demonstration received full marks (20/20). Figure B.1 retains the required screenshot of tracking the setpoint. At 86.10 s it displays reference 5.033 m, height 4.897 m, command 2.498 V and sensor 4.553 V. The displayed error is 0.137 m; subtraction of rounded heights gives 0.136 m.

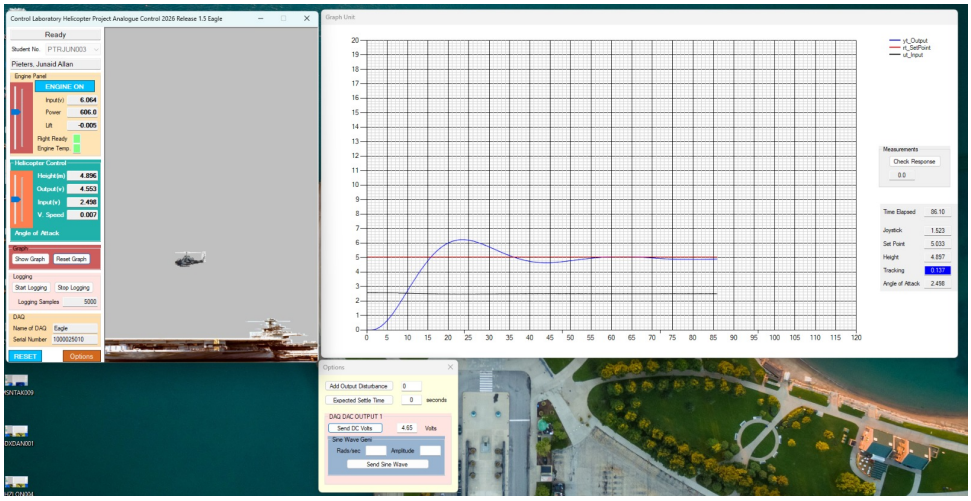


Figure B.1: Original controller tracking screenshot required by the lab brief. The numerical response is replotted in Figure 4.1.

Figure B.2 is the original supplied Simulink plot, preserved for comparison with the original model in Figure 3.2.

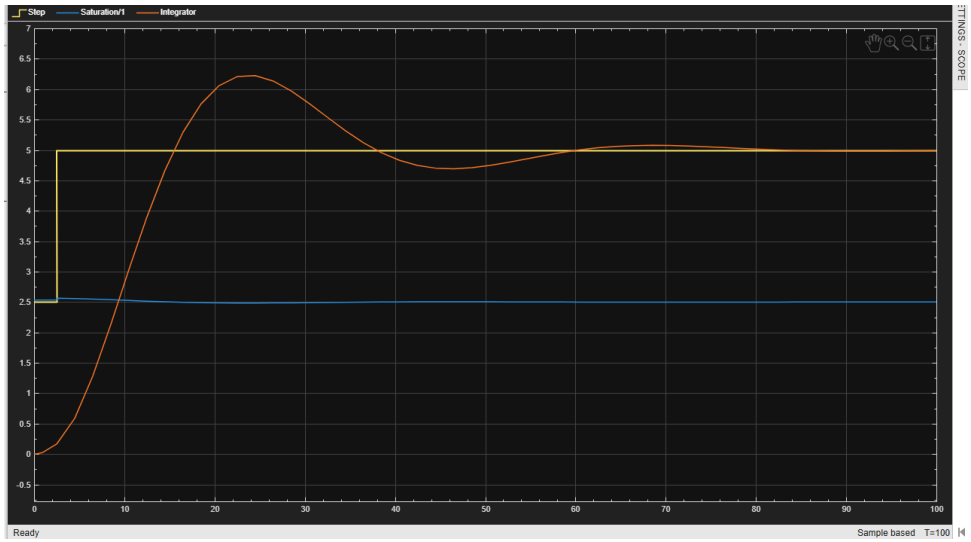
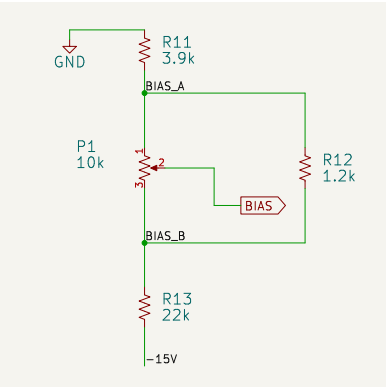


Figure B.2: Original Simulink scope: reference (yellow), command (blue) and altitude (orange). The matched-reference comparison is Figure 3.3.

Digitised values are approximate: Simulink altitude resolution is 0.0124 m/pixel, while the demo plot resolution is 0.0417 m/pixel. Source images, processed CSVs and Simulink reproduction scripts remain accessible through the linked repository folders.

Appendix C: Supporting Calculations

The following enlarged KiCad view shows the bias source feeding U3. R_{11} connects to ground; R_{13} connects to -15 V. The pot wiper is loaded through R_9, so its unloaded voltage alone does not determine hover output.



DAQ connections: DAC-1 reference, DAC-0 sensor, ADC-1 controller output, ADC-2 sensor monitor. All grounds are common. Op-amp pin 7 is +15 V and pin 4 is −15 V.

U1: $a_f = R_4/R_3$, $a_r = (1+a_f)R_2/(R_1+R_2)$; nominally both are 0.01. U2: $g_2 = 1+R_5/R_6 = 1.47$. U3: $g_3 = (1+R_{10}/(R_9+R_{\text{th}}))R_8/(R_7+R_8)$.

Figure C.1: Bias network and DAQ mapping supporting the enlarged signal stages in Figure 3.4.

For the unloaded network, $R_P = 10 \parallel 1.2 = 1.071429$ kΩ and $I = -15/(3.9 + R_P + 22) = -0.556143$ mA. Thus $V_A = -2.168962$ V, $V_B = -2.764831$ V and $V_{\text{th}} = V_A + W(V_B - V_A)$. Solving the loaded hover condition gives $W = 0.783081$, $R_{\text{th}} = 5.423001$ kΩ and $V_{\text{th}} = -2.635575$ V:

$$u_0 = -\frac{100}{100 + 5.423001}(-2.635575) = 2.500000\,\text{V}, \tag{C.1}$$

$$g_3 = \frac{1 + 100/(100 + 5.423001)}{2} = 0.974280, \tag{C.2}$$

$$K_p = 0.01(1.47)(0.974280) = 0.014321913. \tag{C.3}$$

The reported 2.565 V potentiometer adjustment is an observation at an unlogged node/time. A permanent zero-error bias of 2.565 V instead of 2.5 V would predict $\delta y = 0.065/(0.01432 \times 0.93) = 4.88076$ m, so it is not silently substituted for theoretical hover. The isolated 5 m step initially commands 2.566588 V; the matched Simulink sequence reaches 2.564405 V.

Independent resistor errors change $K_f = a_f g_2 g_3$ and $y_\infty/x_\infty = a_r/a_f$. The 2,048 resistor/drag corners give 48.79% maximum error and 38.95% overshoot. For example, $a_f = 0.9/110$ and $a_r = (1 + a_f)1.1/(90 + 1.1)$ give $a_r/a_f = 1.487864$. Matched ratios and two 680 Ω U1 resistors give $K_p = 0.00973890$ and 23.44% worst scalar overshoot; physical retesting is required. At the selected root $s_d = -0.0617284 + j0.1402123$, pole angles are 113.7615° and 66.2385° (sum 180°). The magnitude condition gives

$$K_p = \frac{|s_d||s_d + 0.1234568|}{1.6389585} = \frac{(0.1531988)^2}{1.6389585} = 0.01432. \tag{C.4}$$

Extended calculations, Python code, MATLAB root locus, Simulink tests and reproduction instructions are supplied outside the report.